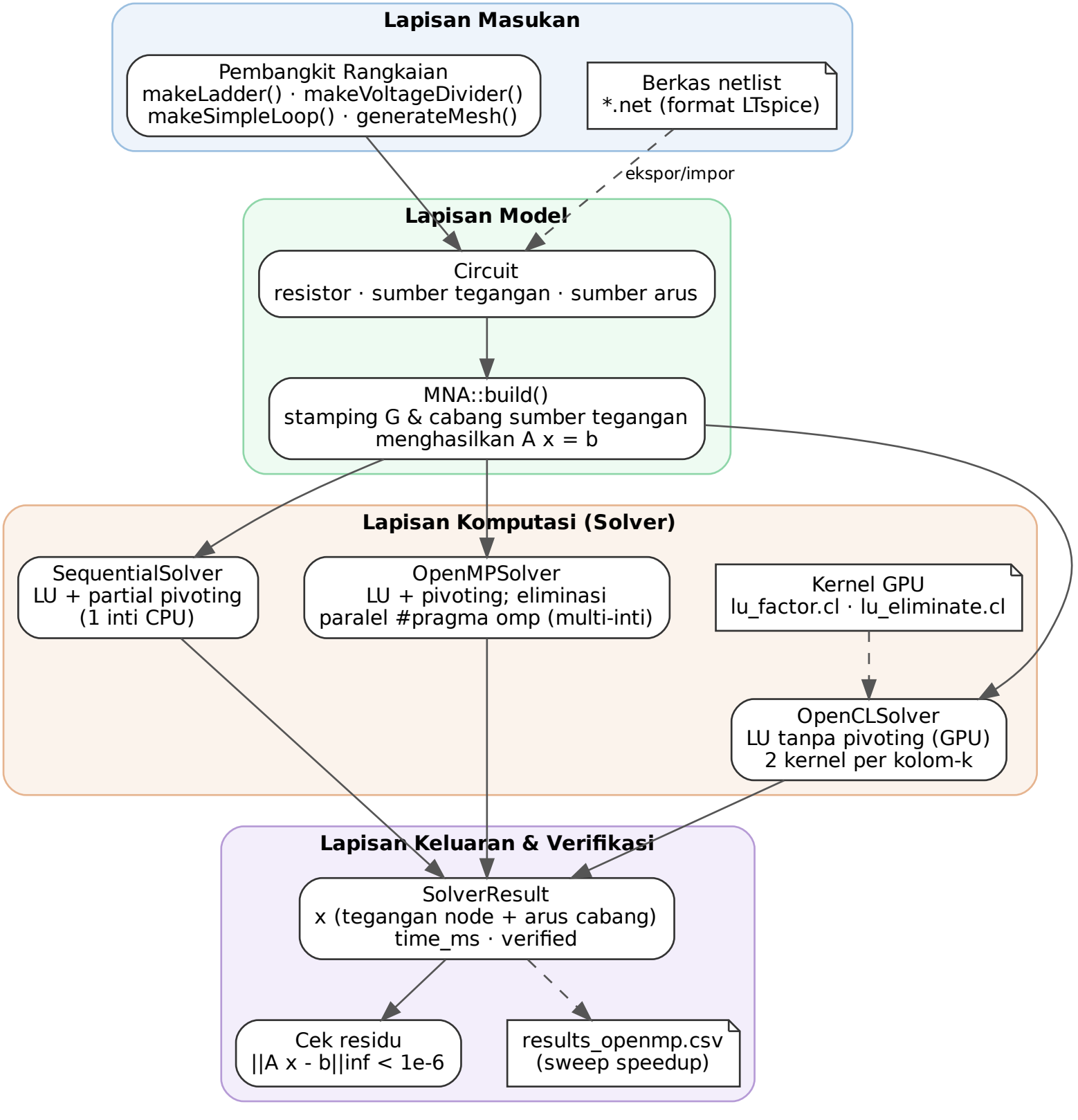
Lapisan Masukan
Pembangkit Rangkaian
makeLadder() · makeVoltageDivider()
makeSimpleLoop() · generateMesh()
Berkas netlist
*.net (format LTspice)
ekspor/impor
Lapisan Model
Circuit
resistor · sumber tegangan · sumber arus
MNA::build()
stamping G & cabang sumber tegangan
menghasilkan A x = b
Lapisan Komputasi (Solver)
SequentialSolver
LU + partial pivoting
(1 inti CPU)
OpenMPSolver
LU + pivoting; eliminasi
paralel #pragma omp (multi-inti)
Kernel GPU
lu_factor.cl · lu_eliminate.cl
OpenCLSolver
LU tanpa pivoting (GPU)
2 kernel per kolom-k
Lapisan Keluaran & Verifikasi
SolverResult
x (tegangan node + arus cabang)
time_ms · verified
Cek residu
||A x - b||inf < 1e-6
results_openmp.csv
(sweep speedup)